Vector space of bit vectors

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Abstract

The aim of this article is to document all properties of the vector space of size n of bit vectors $\mathbb{F}_2^n$ conscisly in one place. We include the properties usually only documented for the vector spaces $\mathbb{Q}^n, \mathbb{R}^n, \mathbb{C}^n$, like base, span, dimension, norm, scalar product and inner sum. For the properties norm and scalar product we provide definitions with similar properties satisfying the axioms.

1 Introduction

Bit vectors are frequently used in computer science. They are used for integers, combinatorial algorithms, coding theory, and for logical and arithmetic operations [9, 7]. A set of bit vectors of the same length forms a vector space. Information of all aspects of the vector space of bit vectors are scattered over multiple documents. This article is a collection of the properties of the vector space of bit vectors of the same length.

Bits are based on the finite set of integers of order 2. This set $\{0, 1\}$ with the operations addition and multiplication modulo 2 satisfies the axioms of a field. This finite field is named Galois Field [10, 17][1] $GF(2) = \mathbb{F}_2$. Over this field there is the vector space $\mathbb{F}_2^n$.

The Go package `gf2vs` [12] implements data types and functions for the vector space of bit vectors. This article is a supplementary documentation to the Go package.

2 Field $\mathbb{F}_2$

2.1 Supporting Set

In computer science bits are modeled as the set $\{0, 1\}$. This is the set $\mathbb{Z}_2 \subset \mathbb{N} \subset \mathbb{Z} \subset \mathbb{R}$. This set is equivalent to the set of representatives of the cyclic group $\mathbb{Z}/2\mathbb{Z}$ of order 2. In logic we have the boolean values False $F = 0$ and True $T = 1$ [8].

The representatives of the cyclic set $\mathbb{Z}/2\mathbb{Z}$ with operations modulo 2 satisfy the field axioms. We use the notation $\mathbb{F}_2$ for the field. It is equivalent to the Galois Field $GF(2)$.

2.2 Group Operations

The operations of $\mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z}$ are defined modulo 2; see [3, ch. 2.2.6] and [7].

The operations addition and multiplication of the field $\mathbb{F}_2$ satisfy the group axioms [3, 10, 18], both operations are commutative.

[1] We cite Wikipedia for reused wordings.

Similarly the operations addition and multiplication of the field $\mathbb{R}$ satisfy the group axioms [3], both operations are commutative.

Remark: Please note the different definition of the addition in $\mathbb{F}_2$ and $\mathbb{R}$.

For reference we give here only the operations for $\mathbb{F}_2$.

2.2.1 Addition

The addition is named exclusive disjunction in logic and XOR in computer science [8, 16]. The definition of addition is given in equation (1) obeying $(1 + 1) \bmod 2 = 0$.

$$\begin{aligned} + : \mathbb{F}_2 \times \mathbb{F}_2 \longrightarrow \mathbb{F}_2, \quad (a, b) &\mapsto (a + b) \bmod 2, \\ 0 + 0 = 0, \quad 0 + 1 = 1, \quad 1 + 0 &= 1, \quad 1 + 1 = 0. \end{aligned} \tag{1}$$

The group axioms [3, ch. 2.2.8] for the Group $G = \mathbb{F}_2$ and the operation addition are satisfied:

Associativity
$\forall a, b, c \in G : (a + b) + c = a + (b + c)$.

Identity element $e = 0$
$\exists e \in G, \forall a \in G : e + a = a$ and $a + e = a$, $e = 0$, e is unique.

Inverse element $(-a) = a$
$\forall a \in G \ \ \exists (-a) \in G : a + (-a) = e$ and $(-a) + a = e$, e identity element, $(-a) = a$ is unique for each a. The elements of the set $\{0, 1\}$ are their own inverses, see equation (1).

Commutativity
$a + b = b + a$.

So $\mathbb{F}_2$ with the operation addition is an abelian group.

2.2.2 Multiplication

The multiplication is named conjunction in logic and AND in computer science [8, 20]. The multiplication is identically defined as in $\mathbb{Z}$.

$$\begin{aligned} \cdot : \mathbb{F}_2 \times \mathbb{F}_2 \longrightarrow \mathbb{F}_2, \quad (a, b) &\mapsto (a \cdot b) \bmod 2, \\ 0 \cdot 0 = 0, \quad 0 \cdot 1 = 0, \quad 1 \cdot 0 &= 0, \quad 1 \cdot 1 = 1. \end{aligned} \tag{2}$$

We may use the notation ab instead of $a \cdot b$, omitting the multiplication sign if there is no ambiguity.

The group axioms for the Group $G = \mathbb{F}_2$ and the operation multiplication are satisfied:

Associativity
$\forall a, b, c \in G : (a \cdot b) \cdot c = a \cdot (b \cdot c)$.

Identity element $e = 1$
$\exists e \in G, \forall a \in G : e \cdot a = a$ and $a \cdot e = a$, $e = 1$, e is unique.

Inverse element a^{-1}
$\forall a \in G, a \neq 0, \ \ \exists a^{-1} \in G : a \cdot a^{-1} = e$ and $a^{-1} \cdot a = e$, e is the identity element; the only invertible element is 1, hence $a^{-1} = 1$ for $a = 1$.

Commutativity
$a \cdot b = b \cdot a$.

So $\mathbb{F}_2$ with the operation multiplication is an abelian group.

2.3 Mappings

In addition to the group operations we define some mappings, applicable to $\mathbb{F}_2$.

2.3.1 Complement

$$\begin{aligned} \neg : \mathbb{F}_2 \to \mathbb{F}_2, \quad \neg x = 1 + x, \\ \neg 0 = 1, \quad \neg 1 = 0. \end{aligned} \tag{3}$$

2.3.2 Disjunction

In boolean algebra we have the operation disjunction, named OR in computer science [8, 21].

$$\begin{aligned} \vee : \mathbb{F}_2 \times \mathbb{F}_2 \to \mathbb{F}_2, \quad (a, b) \mapsto a \vee b, \\ 0 \vee 0 = 0, \quad 0 \vee 1 = 1, \quad 1 \vee 0 = 1, \quad 1 \vee 1 = 1. \end{aligned} \tag{4}$$

The operation $\vee$ does not satisfy the group axioms; there is no inverse element.

2.3.3 Absolute Value

Absolute value $|\cdot|$ maps an element x of $\mathbb{F}_2$ to $\mathbb{R}$:

$$\begin{aligned} |\cdot| : \mathbb{F}_2 \to \mathbb{R}, \quad x \mapsto y, \\ |0| = 0, \quad |1| = 1. \end{aligned} \tag{5}$$

After application of this mapping, we can use the classical definition of the addition as in $\mathbb{R}$.

2.4 Field Axioms

The set $\mathbb{F}_2$ with the operations addition and multiplication satisfies the field axioms [3, 7]. We use K as symbol for any field satisfying the field axioms.

K1 K with the addition + is an abelian group.

K2 $K^* := K \setminus \{0\}$ with the multiplication $\cdot$ for every element of K^* is an abelian group.

K3 distributive property [2, 14] is satisfied $\forall a, b, c \in K$

$$\begin{aligned} a \cdot (b + c) = a \cdot b + a \cdot c, \\ (a + b) \cdot c = a \cdot c + b \cdot c. \end{aligned} \tag{6}$$

Obviously $\mathbb{F}_2$ satisfies the field axioms.

3 Vector Space $\mathbb{F}_2^n$

3.1 Vectors

We define a bit vector x of size n as a tuple (x_i) of values $x_i \in \mathbb{F}_2$:

$$x := (x_i) := (x_1, x_2, \ldots, x_n), \quad \forall x_i \in \mathbb{F}_2, \tag{7}$$

and we define the set of all bit vectors of size n see [3, ch 2.4.1] as the vector space $\mathbb{F}_2^n$:

$$\mathbb{F}_2^n := \{x = (x_1, \dots, x_n) : x_i \in \mathbb{F}_2\}. \tag{8}$$

In addition we define two distinguished constant elements of $\mathbb{F}_2^n$:

Zero $\mathbf{0}$, zero vector all components are 0, identity element of vector addition equation (13).

Ones $\mathbf{1}$, ones vector all components are 1, identity element of vector multiplication.

3.2 Mapping of Integers to Bit Vectors

All vectors in a vector space have the same length n. Obviously we do not have a vector space if $n = 0$.

Bit vectors are representatives of unsigned integers $a_i \in \mathbb{N}$. For mapping of a set of unsigned integers we compute the order n of thei target vector space from the maximum of the $\max(a_i) > 0$, using equation (9). We restrict the vector space to positive integers. If $\max(a_i) \leq 0$ the a_i cannot be represented with the vector space as defined in this article. Regarding the special considerations for representation of integers $a_i < 0$ as bit vectors see [9].

$$n = \lfloor \log_2(\max(a_i)) \rfloor + 1. \tag{9}$$

Knowing n we can use equation (10) to map $a \in \mathbb{N}, 0 \leq a \leq n$.

$$\varphi : \mathbb{N} \to \mathbb{F}_2^n, \quad x = \left(\lfloor \tfrac{a}{2^0} \rfloor \bmod 2,\ \lfloor \tfrac{a}{2^1} \rfloor \bmod 2,\ \dots,\ \lfloor \tfrac{a}{2^{n-1}} \rfloor \bmod 2 \right). \tag{10}$$

The inverse mapping φ^{-1} is given by equation (11)

$$\varphi^{-1} : \mathbb{F}_2^n \to \mathbb{N}, \quad a = \textstyle\sum_{i=1}^n x_i 2^{i-1}. \tag{11}$$

In the programming language Go, as in most programming languages these mapping are used when parsing integer strings and formatting integers. Internally integers are stored as bit vectors compatible with the hardware word size of the target computer for the programming language. This limits on most hardware architectures the vector size to a power of 2 of the vectors used to represent integers. The Go package `gf2vs` supports vector spaces of arbitrary length, up to 64 bit.

3.3 Operations on Bit Vectors

On bit vectors we have mappingis defining the operations of the vector space and additional mappings ususally not present in definitions of a vector space.

3.3.1 Mappings of Bit Vectors

Equation (12) defines the mapping of bit vectors.

$$\left.\begin{array}{llcll} \text{Complement} & \sim : \mathbb{F}_2^n \to \mathbb{F}_2^n, & \sim x = y & \Leftrightarrow & \neg x_i = y_i\ , \\ \text{Absolut value} & |\cdot| : \mathbb{F}_2^n \to \mathbb{R}^n, & |x| = y & \Leftrightarrow & |x_i| = |y_i|, \end{array}\right\} i = 1, \dots, n. \tag{12}$$

We define the mapping $|\cdot|$ for formal mapping of a bit vector from $\mathbb{F}_2^n$ to $\mathbb{R}^n$.

For the constants we have: $\sim \mathbf{0} = \mathbf{1}$ and $\sim \mathbf{1} = \mathbf{0}$.

3.3.2 Operations on Bit Vectors

Equation (13) defines bit-wise operations on the bit vectors x, y, z see [3, ch. 2.4.1] and [9, (1), (2), (3)]. As we have a vector space over a field the vector operations defined bit-wise inherit the group properties of the field $\mathbb{F}_2$.

$$\left.\begin{array}{llrcl} \text{Addition, XOR} & + : \mathbb{F}_2^n \times \mathbb{F}_2^n \to \mathbb{F}_2^n, & x + y = z & \Leftrightarrow & x_i + y_i = z_i, \\ \text{Multiplication, AND} & \cdot : \mathbb{F}_2^n \times \mathbb{F}_2^n \to \mathbb{F}_2^n, & x \cdot y = z & \Leftrightarrow & x_i \cdot y_i = z_i, \\ \text{Disjunction, OR} & | : \mathbb{F}_2^n \times \mathbb{F}_2^n \to \mathbb{F}_2^n, & x|y = z & \Leftrightarrow & x_i \vee y_i = z_i, \\ \text{Scalar multiplication} & \cdot : \mathbb{F}_2 \times \mathbb{F}_2^n \to \mathbb{F}_2^n, & \lambda \cdot x = y & \Leftrightarrow & \lambda \cdot x_i = y_i, \end{array}\right\} i = 1, \dots, n. \tag{13}$$

We adopt the main identities from [9, (4), . . . , (14)] for bit vectors of size n here:

$$x + y = y + x, \;\; x \cdot y = y \cdot x, \;\; x|y = y|x : \qquad \text{commutativity;} \tag{14}$$

$$(x + y) + z = x + (y + z), \;\; (x \cdot y) \cdot z = x \cdot (y \cdot z), \;\; (x|y)|z = x|(y|z) : \qquad \text{associativity;} \tag{15}$$

$$(x + y) \cdot z = (x \cdot z) + (y \cdot z) : \qquad \text{distributivity AND over XOR;} \tag{16}$$

$$(x \cdot y)|z = (x|z) \cdot (y|z), \;\; (x|y) \cdot z = (x \cdot z)|(y \cdot z) : \qquad \text{mutual distributivity AND, OR;} \tag{17}$$

$$(x \cdot y) + (x|y) = x + y; \tag{18}$$

$$(x \cdot y)|x = x, \;\; (x|y) \cdot x = x : \qquad \text{idempotence;} \tag{19}$$

$$x + \mathbf{0} = x, \;\; x \cdot \mathbf{0} = \mathbf{0}, \;\; x|\mathbf{0} = x; \tag{20}$$

$$x + \mathbf{1} = \sim x, \;\; x \cdot \mathbf{1} = x, \;\; x|\mathbf{1} = \mathbf{1}; \tag{21}$$

$$x + x = \mathbf{0}, \;\; x \cdot x = x, \;\; x|x = x; \tag{22}$$

$$x + (\sim x) = \mathbf{1}, \;\; x \cdot (\sim x) = \mathbf{0}, \;\; x|(\sim x) = \mathbf{1}; \tag{23}$$

$$\sim (x + y) = (\sim x) + y = x + (\sim y); \tag{24}$$

$$\sim (x \cdot y) = (\sim x)|(\sim y), \;\; \sim (x|y) = (\sim x) \cdot (\sim y) : \qquad \text{De Morgan.} \tag{25}$$

3.4 Axioms of Vector Space

The set $\mathbb{F}_2^n$ with the binary operation of vector addition $+$ and the binary function of scalar multiplication $\cdot$, as given in equation (13), define a vector space see [3, 24]. We repeat here the axioms of a vector space and leave the proof to the reader.

Associativity of vector addition
$u + (v + w) = (u + v) + w, \;\; \forall u, v, w \in \mathbb{F}_2^n.$

Commutativity of vector addition
$u + v = v + u, \;\; \forall u, v \in \mathbb{F}_2^n.$

Identity element of vector addition
$\exists \mathbf{0} \in \mathbb{F}_2^n : v + \mathbf{0} = v, \;\; \forall v \in \mathbb{F}_2^n.$

Inverse elements of vector addition
$\forall v \in \mathbb{F}_2^n \;\; \exists -v \in \mathbb{F}_2^n : v + (-v) = \mathbf{0}$, and $-v = v$, i.e. each vector is its own additive inverse, see equation (1).

Compatibility of scalar multiplication with field multiplication
$\lambda(\eta v) = (\lambda\eta)v, \;\; \lambda, \eta \in \mathbb{F}_2, \;\; v \in \mathbb{F}_2^n.$

Identity element of scalar multiplication
$1v = v, \;\; 1 \in \mathbb{F}_2, \;\; v \in \mathbb{F}_2^n$, where 1 is the multiplicative identity of $\mathbb{F}_2$.

Distributivity of scalar multiplication with respect to vector addition
$\lambda(u + v) = \lambda u + \lambda v, \;\; \lambda \in \mathbb{F}_2, \;\; u, v \in \mathbb{F}_2^n.$

Distributivity of scalar multiplication with respect to field addition
$(\lambda + \eta)v = \lambda v + \eta v, \;\; \lambda, \eta \in \mathbb{F}_2, \;\; v \in \mathbb{F}_2^n.$

The axioms of a vector space are satisfied for $\mathbb{F}_2^n$

3.5 Vector Space Base

For the vector space $\mathbb{F}_2^n$ we give here the definition of the base, the norm and the scalar product as implemented in the Go package.

3.5.1 Unit Vector

We define the unit vectors

$$e_i = (0, \dots 0, 1, 0, \dots, 0), \quad i = 1, \dots, n, \tag{26}$$

of the vector space as the vectors where the ith element is 1 and all other elements are 0.

$$e_i = (x_k), \;\; e_i \in \mathbb{F}^n, \;\; x_k \in \mathbb{F}, \;\; x_k = \begin{cases} 1, & k = i, \quad \text{identity element of multiplication,} \\ 0, & k \neq i, \quad \text{identity element of addition.} \end{cases} \tag{27}$$

Please note the e_i are linearly independent.

In the vector space $\mathbb{F}_2^n$ each vector x satisfies the equation (28) iff x is a unit vector [1, 6]

$$(x \neq \mathbf{0}) \wedge ((x \cdot (x + e_1))) = \mathbf{0}. \tag{28}$$

The equation (28) is used in the function `x.IsBaseVector()` of `gf2vs.go`. The algorithm is the fastest test to verify a vector is a unit vector of $\mathbb{F}_2^n$ for arbitrary n.

We observe the identity elements are identical for the fields and the unit vectors are identical for all vector spaces over a field K.

3.5.2 Index

We call $i = 1, \dots, n$ the index of the unit vector e_i in the base. In the software package `gf2vs.go` we use the Go library function `math/bits.Len(x uint)` for computing the index of a given unit vector. The function `math/bits.Len(x)` computes the number of bits required to represent an unsigned integer. This is an optimized implementation of equation (9).

3.5.3 Generating System

We define the subset $\mathbb{E} := \{e_i\}$, $\mathbb{E} \subset \mathbb{F}_2^n$, of unit vectors e_i. The subset $\mathbb{E}$ forms a generating system. Each vector v of $\mathbb{F}_2^n$ is a linear combination of scalars a_i and the e_i:

$$v = \sum_{i=1}^{n} a_i e_i, \;\; a_i \in \mathbb{F}_2, \;\; e_i \in \mathbb{E}, \quad \forall v \in \mathbb{F}_2^n. \tag{29}$$

For the vector space $\mathbb{F}_2^n$ the addition is modulo 2.

Equation (29) is used equally for each vector space on any field K using the operation addition as defined for the field K and the 1 the identity element of the operation multiplication.

Thus the subset $\mathbb{E}$ spans $\mathbb{F}_2^n$. In this vector space it is one spanning set, and the decomposition of a vector v into a linear combination of unit vectors e_i is unique.

3.5.4 Base

The subset $\mathbb{E}$ is one base of the vector space $\mathbb{F}_2^n$. As it is a base of every vector space over a field K.

For the vector space $\mathbb{F}_2^n$ the sum of the base vectors is $\mathbf{1}$ the ones vector..

$$\sum_{i=1}^{n} e_i = \mathbf{1} \tag{30}$$

3.5.5 Norm via Hamming Weight

The addition in the field $\mathbb{F}_2$ is modulo 2. Hence each sum in $\mathbb{F}_2$ evaluates to either 0 or 1. In particular, for $x \in \mathbb{F}_2^n$ we have

$$\left(\sum_{i=1}^{n} x_i\right) \bmod 2 = \begin{cases} 1, & |\{i \in \{1,\dots,n\} : x_i = 1\}| \text{ is odd,} \\ 0, & |\{i \in \{1,\dots,n\} : x_i = 1\}| \text{ is even.} \end{cases} \tag{31}$$

From this it follows that we cannot directly use the usual norm definitions (as over $\mathbb{R}$) to measure vector length in $\mathbb{F}_2^n$.

In coding theory [5, 19] the *Hamming weight* w_H (number of 1-entries) of $x \in \{0,1\}^n$ is defined as:

$$w_H : \mathbb{F}_2^n \to \mathbb{N}, \quad w_H(x) = |\{i \in \{1,\dots,n\} : x_i = 1\}|, \tag{32}$$

and the associated *Hamming distance* d_H is:

$$d_H : \mathbb{F}_2^n \times \mathbb{F}_2^n \to \mathbb{N}, \quad d_H(x,y) = w_H(x-y). \tag{33}$$

If we first apply operation $|\ |$ we map a vector from $\mathbb{F}_2^n \to \mathbb{R}^n$, by embedding $\mathbb{F}_2^n = \{0,1\}^n \subset \mathbb{R}^n$. On vectors $x = (x_i)$ in $\mathbb{R}^n$ norms are defined and we get:

$$\|x\|_1 = \sum_{i=1}^{n} |x_i| = w_H(x). \tag{34}$$

In $\mathbb{R}^n$ $\|x\|_1$ is named ℓ_1-norm see [4, 13]. Obviously it equals the Hamming weight on bit vectors. This norm is sometimes called absolute-value norm [22]. The value of the norm of a vector of $\mathbb{F}_2^n$ is an element of the set $\{0,1,\dots,n\} \subset \mathbb{N} \subset \mathbb{Z} \subset \mathbb{R}$.

In the programming languages C the function is named popcount() and in the language Go it is the function `OnesCount(x uint) uint` in package `math/bits`.

This norm satisfies the axioms of a norm:

Subadditivity / Triangle inequality
$w_H(x+y) \leq w_H(x) + w_H(y) \quad \forall x, y \in \mathbb{F}_2^n,$

Absolute homogeneity
$w_H(s \cdot x) = s \cdot w_H(x) \quad \forall s \in \mathbb{F}_2, \ x \in \mathbb{F}_2^n,$

Positive definiteness
$w_H(x) \geq 0$, and $w_H(x) = 0 \Leftrightarrow x = \mathbf{0}$.

Please note the norm of the unit vectors

$$\|e_i\|_1 = 1, \quad \forall i \in \{1,\dots n\}. \tag{35}$$

These are the only vectors with norm 1 in the vector space.

3.5.6 Scalar Product

We define the scalar product, dot product [3, 15] or inner product of two vectors as given in equation (36):

$$\langle , \rangle : \mathbb{F}_2^n \times \mathbb{F}_2^n \to \mathbb{R}, \quad \langle x, y \rangle = \textstyle\sum_{i=1}^{n} |x_i \cdot y_i| = \|x \cdot y\|_1 = w_H(x \cdot y). \tag{36}$$

The obtained value equals the standard scalar product of $x, y \in \{0,1\}^n \subset \mathbb{R}^n$

$$\langle , \rangle : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}, \quad \langle x, y\rangle = \textstyle\sum_i^n x_i \cdot y_i. \tag{37}$$

By computation we can easily verify the properties of the scalar product, for $x, y \in \mathbb{F}_2^n$ and $\lambda \in \mathbb{F}_2$:

Distributivity

$$\begin{aligned}
&\langle x + x', y\rangle = \langle x, y\rangle + \langle x', y\rangle,\\
&\langle x, y + y'\rangle = \langle x, y\rangle + \langle x, y'\rangle,\\
&\langle \lambda x, y\rangle = \lambda\langle x, y\rangle,\\
&\langle x, \lambda y\rangle = \lambda\langle x, y\rangle.
\end{aligned}$$

Commutativity

$$\langle x, y\rangle = \langle y, x\rangle.$$

Positive definiteness

$$\langle x, x\rangle \quad \geq 0, \text{ and } \langle x, x\rangle = 0 \Leftrightarrow x = \mathbf{0}.$$

Orthogonality

$$\begin{aligned}
&\langle x, y\rangle = 0, \text{ we say the two vectors are orthogonal.}\\
&\langle x, \sim x\rangle = 0, \text{ a vector is orthogonal to its complement.}
\end{aligned}$$

In coding theory [11] the orthogonal bit vector is called the dual code.

3.5.7 Orthonormal Base

With the properties of the norm and the scalar product it follows $\mathbb{E}$ is an orthonormal base, and this is the only orthonormal base of $\mathbb{F}_2^n$, as the unit vectors are the only vectors in $\mathbb{F}_2^n$ with a length of 1 in the vector space, see equation (35).

3.6 Sets of Bit Vectors

3.7 Subspaces

3.7.1 Definition

As for other vector spaces, we can define subspaces. A subset U of $\mathbb{F}_2^n$ is a subspace iff U satisfies the three conditions [2, Theorem 1.34].

additive identity
: $\mathbf{0} \in U$.

closed under addition
: $u, w \in U \Rightarrow u + w \in U$.

closed under scalar multiplication
: $a \in \mathbb{F}, u \in U \Rightarrow au \in U$.

For $\mathbb{F}_2^n$ every subset of $\mathbb{E}$ spans a subspace. The conditions can be easily verified.

3.7.2 Direct Sum of Subspaces

Every subspace U of $\mathbb{F}_2^n$ is part of a direct sum equal to V as described by [2, Theorem 2.33]. As per definition $\mathbb{F}_2^n$ is a finite-dimensional vector space. Let U be a subspace of $\mathbb{F}_2^n$. Then there exists a subspace W of $\mathbb{F}_2^n$ such that $V = U + W$.

Let B_V be the canonical base of V so $B_V \subseteq \mathbb{E}$ then $B_W = \mathbb{E} \setminus B$ is a base and we have $B_V \cup B_W = \mathbb{E}$. Let W be the subspace spanned by B_W, then $V \cap W = \{\mathbf{0}\}$ and $V + W = \mathbb{F}_2^n$. For the proof see [2].

Suppose we have two disjunctive sets of vectors $V, U, V \cap U = \emptyset$ of vector space $\mathbb{F}_2^n$, each with corresponding bases B_V, B_U. The bases do not necessarily have to be disjoint.

If the span $V = B_V$ of one subset V, then V is a subspace of $\mathbb{F}_2^n$. In this situation we can create a subspace W out of U so that the direct sum $U + W = \mathbb{F}_2^n$. This is an application of the Steinitz exchange lemma [3, 23].

Assume without loss of generality $B_V = \{e_1, ..., e_k\}$ and $U = \{u_1, ...u_l\}$. We compute the base B_W of W with equation (38)

$$B_W = \mathbb{E} \setminus B_V \tag{38}$$

With the base B_W we can restrict the set of vectors U to a set of vectors $W = \{w_1, ..., w_l\}$ making W a subspace with base B_W simply by filtering the bit vectors of U with a bit mask m

$$m = \sum_{i=1}^{l} e_i, \quad e_i \in B_w, \tag{39}$$

using equation (40) we get

$$w_i = u_i \cdot m, \quad \forall u_i \in U. \tag{40}$$

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